Huawei Access Node JAY-ACC-13 Routing Protocol Flap SOP

Node: JAY-ACC-13 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7816

Description

This standard operating procedure outlines the steps to diagnose and resolve a Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure on Huawei Access Node JAY-ACC-13 at the Jayanagar Metro Site. The failure is characterized by Alarm Code ALARM_7816 and Severity Critical. The root cause of this failure is attributed to the dynamic nature of routing protocols, particularly BGP, OSPF, and ISIS, which can lead to rapid route churn and adjacency loss due to various factors such as misconfiguration, interface issues, or external network disruptions. In the context of Huawei equipment, this can manifest as a result of incorrect configuration of BGP, OSPF, or ISIS, or due to hardware issues such as interface flapping or module failures. The diagnostic commands and resolution steps outlined in this SOP are designed to isolate and correct these issues, thereby preventing further network disruptions and ensuring stable routing protocol operation.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Jayanagar Metro Site to perform physical checks and troubleshooting.

- Verify the status of redundant links to ensure they are operational and not contributing to the failure.

- Check the system logs for any recent errors or anomalies related to BGP, OSPF, or ISIS.

Potential Root Causes

Cause	Probability
Incorrect BGP configuration	High
OSPf adjacency loss due to interface issues	Medium
ISIS LSP flooding due to hardware failure	Low

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: showing BGP process status, Description: This command displays the BGP process status, including the number of routes, peers, and any errors. It helps to identify if there are any issues with the BGP configuration or operation.

Command: display ospf neighbor, Expected Output: showing OSPF neighbor status, Description: This command displays the OSPF neighbor status, including the number of neighbors, their IP addresses, and the state of adjacency. It helps to identify if there are any issues with OSPF adjacency or neighbor establishment.
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```
Command: display isis lsp, Expected Output: showing ISIS LSP status, Description: This command displays the ISIS LSP status, including the number of LSPs, their sequence numbers, and the state of flooding. It helps to identify if there are any issues with ISIS LSP flooding or hardware failures.
```

Resolution Steps

Step 1: Step: Correct BGP configuration errors, Command: undo bgp [process-id] [neighbor ip-address], Description: This command corrects any BGP configuration errors, such as incorrect neighbor IP addresses or incorrect BGP parameters.

Step 2: Step: Restore OSPF adjacency, Command: undo ospf [process-id] [neighbor ip-address] shutdown, Description: This command restores OSPF adjacency by removing the shutdown command and allowing the OSPF neighbor to re-establish adjacency.

Step 3: Step: Clear ISIS LSP flooding, Command: clear isis lsp [sequence-number], Description: This command clears ISIS LSP flooding by resetting the sequence number and allowing the LSPs to re-flood.

Post-Check Verification

```
Command: display bgp process, Expected Output: BGP process status showing no errors, Description: This command verifies that the BGP process is operating correctly, with no errors or anomalies.

Command: display ospf neighbor, Expected Output: OSPF neighbor status showing adjacency established, Description: This command verifies that the OSPF neighbor is established and adjacency is up.

Command: display isis lsp, Expected Output: ISIS LSP status showing no flooding, Description: This command verifies that there is no ISIS LSP flooding and the LSPs are operating correctly.
```

Roll-back Procedures

- Step: Restore original BGP configuration, Command: undo undo bgp [process-id] [neighbor ip-address], Description: This command restores the original BGP configuration, including any changes made during the resolution process.

- Step: Restore original OSPF configuration, Command: undo undo ospf [process-id] [neighbor ip-address] shutdown, Description: This command restores the original OSPF configuration, including any changes made during the resolution process.

- Step: Restore original ISIS configuration, Command: undo undo isis lsp [sequence-number], Description: This command restores the original ISIS configuration, including any changes made during the resolution process.